

Advanced sentence-embedding method considering token importance based on explainable artificial intelligence and text summarization model

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ABSTRACT

Although pretrained language models achieve high performance on various natural language processing tasks, they still require further improvements in the sentence embedding task. Many studies have improved performance in this task using pre-trained language models and contrastive learning, but these approaches are limited because they are based on naive average pooling and CLS tokens. Therefore, we propose an advanced sentence-embedding method based on weighted pooling that considers token importance. Specifically, the token importance is calculated by combining an explainable artificial-intelligence module with a text summarization model, and the final sentence embedding is derived through weighted pooling token embedding and token importance. Thus, we derive a sentence embedding that considers both the local information of the token embedding and the global information of the entire sentence. Experimental results reveal that our proposed sentence embedding outperforms other models on both text similarity tasks and text classification. Moreover, the proposed method's robustness is verified through the results of an ablation study.

1. Introduction

Pre-trained language models, such as bidirectional encoder representations from transformers (BERT), have achieved great success in various natural language processing (NLP) tasks such as classification, summarization, and translation [1–3]. The primary advantage of a pre-trained language model is that the token embedding, which contains self-attention information, is pretrained to be applicable for various tasks [4–6].

Another important topic in the field of NLP is the derivation of scientific sentence embeddings. Most sentence embeddings are generated based on token embeddings [7]. However, it remains unclear how best to derive a sentence embedding from a pre-trained language model, and the suboptimal performance of such models on sentence embeddings has been widely documented in prior studies [8,9]. Therefore, this study was conducted to develop an advanced sentence embedding system using a pre-trained language model.

The most widespread sentence embedding methods based on pre-trained language models are as follows [10]: (1) those that use a vector of [Classification (CLS)] tokens, which is the first token of the sentence; and (2) those using the mean or max of the contextualized embeddings of all words in a sentence. Although these methods are intuitive, they generally yield low representation performance, as they do not reflect local and global information simultaneously [11]. In other words, the

CLS token alone does not reflect the local information of individual token embeddings, whereas naive pooling methods do not reflect the global information of the entire sentence.

To solve the low representation performance problem, two primary methodologies have been proposed. The first approach utilizes additional elements, such as various types of layer information [12] or new components added to the model [13], whereas the second approach improves the performance of sentence embedding through contrastive learning [14–16]. Although both approaches improve the performance of sentence embedding, they have fundamental limitations because they are still based on CLS tokens and naive pooling methods.

We therefore propose an advanced sentence embedding method that uses explainable artificial intelligence (XAI) to derive each token's importance for use as a combination weight. Specifically, the importance of each token is obtained by learning a separate head for the summary model, and the token embeddings are combined based on this importance weight. Local information is considered using information from token embeddings, while global information is reflected by considering the importance of each token embedding.

Recently, a study has been conducted using the attention score for the combination of token embeddings. However, the approach in question is a task-dependent method rather than a derivation of universal sentence embedding. Thus, to the best of our knowledge,

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our proposed method is the first to calculate the importance of token embedding using XAI and derive a universal sentence embedding using it as a combination weight.

The remainder of this paper is structured as follows: Section 2 discusses various prior studies on sentence embedding approaches using pre-trained language models, and Section 3 describes the proposed method in detail. Section 4 summarizes the data used in the experiments and presents experimental results demonstrating our method's performance. Finally, Section 5 provides concluding remarks and briefly discusses directions for future research.

2. Literature review

In this section, we provide an overview of prior research on sentence embeddings based on pre-trained language models and explain the SHAP method employed in this study.

2.1. Previous studies on sentence embedding based on pre-trained language models

Sentence embedding, which represents the meaning of a sentence in the form of a dense vector, has been extensively studied as a fundamental NLP task. Leveraging the power of pretrained language models in the sentence embedding task has recently become a popular approach [17]. Early pre-trained language-model-based sentence embedding methods input the sentences to BERT and then construct a sentence embedding from the output contextualized word embeddings, using either the (1) [CLS] token embedding or (2) mean or max of the contextualized embedding of all tokens in the sentence. However, these methods are limited in that they cannot reflect the local and global information of a sentence simultaneously. In practice, these approaches have been employed in various studies [18,19], but exhibited suboptimal performance.

To improve the representational performance of sentence embeddings, a study was conducted using additional elements. Specifically, WhitenBERT [12] uses information from both the top and bottom layers in the average pooling process, thereby normalizing sentence embeddings through the whitening technique. As an alternative, speBERT [13] leverages different sentence components, in contrast to previous approaches that only take the embedding vector of the token [CLS] as the sentence embedding. speBERT obtains the final sentence embedding by merging the [CLS] embedding with the component embedding via an aggregator. Multimodal contrastive learning of sentence embeddings (MCSE) [20] was the first attempt to learn sentence embeddings using text with another modality in the form of images. Although MCSE is difficult to generalize because it requires aligned image-text pairs, it has been shown to enhance embedding quality by utilizing additional modalities.

To improve sentence embedding performance using a pre-trained language model, most recently proposed methods are based on contrastive learning. SimCSE [14] incorporates annotated pairs from natural language inference datasets into its contrastive learning framework, using entailment pairs as positives and contradiction pairs as hard negatives. DiffCSE [21] represents an instance of equivariant contrastive learning that generalizes contrastive learning and learns representations that are insensitive to certain types of augmentations and sensitive to other "harmful" types of augmentations. In addition, PromptBERT [17] uses a prompt-based contrastive learning method with template denoising to leverage the power of BERT in unsupervised settings, significantly reducing the gap between supervised and unsupervised performance. Sentence-T5 [22] represents the first exploration of sentence embeddings from text-to-text transformers (T5), using an encoder-decoder architecture and generative span corruption pre-training task. Although many other methods based on contrastive learning [15,23,24] have contributed to improvements in sentence

embedding performance, limitations remain because said methods are fundamentally based on CLS tokens and pooling.

Specifically, existing methods either fail to reflect local information because they are based on CLS tokens, or fail to reflect global information because they are based on naive pooling. We therefore propose an advanced sentence embedding method that simultaneously reflects local information by pooling token embedding information, and global information by weighing the XAI-based tokens in the pooling process.

2.2. Use of SHAP in our proposed method

SHapley Additive exPlanations (SHAP) is one of the most widely used XAI methods [25], which associates optimal credit allocation with local explanations using the classic Shapley values from game theory and their related extensions [25–27]. In machine learning, the Shapley value can be interpreted as the degree to which a specific feature affects the prediction result, as shown in Fig. 1.

It is simple to calculate the Shapley value given a simple model. However, it is much more difficult to calculate the Shapley value when using a complex model such as an ensemble or deep learning model. To interpret a model with a complex structure in SHAP, a simpler approximation of the original model is used. This process is illustrated in the figure below, where a surrogate model g satisfying $g(z') = f(h_x(z'))$ approximates the original model f (see Fig. 2).

Because SHAP is a model-agnostic approach, it can be applied to most classification and regression prediction models. However, in the NLP field, it is most widely used for deriving token sentiment scores in sentiment analysis [28–30].

Another methodology widely used to derive an explainability score is Locally Interpretable Model-agnostic Explanations (LIME) [31]. LIME is also an efficient approach to generate explainability scores, but it fundamentally relies on constructing a surrogate model. Therefore, for complex tasks such as text summarization, effective surrogate model construction can be challenging, leading to a significant potential decrease in the accuracy of explanatory inference. Moreover, LIME assumes linearity even locally, making it difficult to ensure accurate explanatory inference in language models with a high likelihood of non-linearity, even at a local level.

On the contrary, SHAP infers explanatory power not just locally but also from a holistic model perspective, enabling the extraction of explanatory power even in complex models and facilitating its application across various tasks. Consequently, in this study, we have employed the SHAP methodology to derive explainability scores.

3. Method

As mentioned above, our proposed method focuses on the derivation of pre-trained language-model-based sentence embedding while considering the importance of each token. The primary contribution of our proposed model is its ability to reflect the importance of tokens in the sentence-embedding process. Throughout this study, the importance of each token was determined using a text-summarization model. Text summarization is a technique for generating a concise and precise summary of large texts while focusing on sections that convey useful information without losing the overall meaning. Therefore, the text summarization model is highly likely to be learned by focusing on important tokens in the original text.

Our proposed method is a straightforward approach comprised of three major steps: (1) construction of the text summarization model, (2) calculation of token importance based on the text summarization model, and (3) derivation of a sentence embedding reflecting token importance. The proposed method is summarized in Fig. 3.

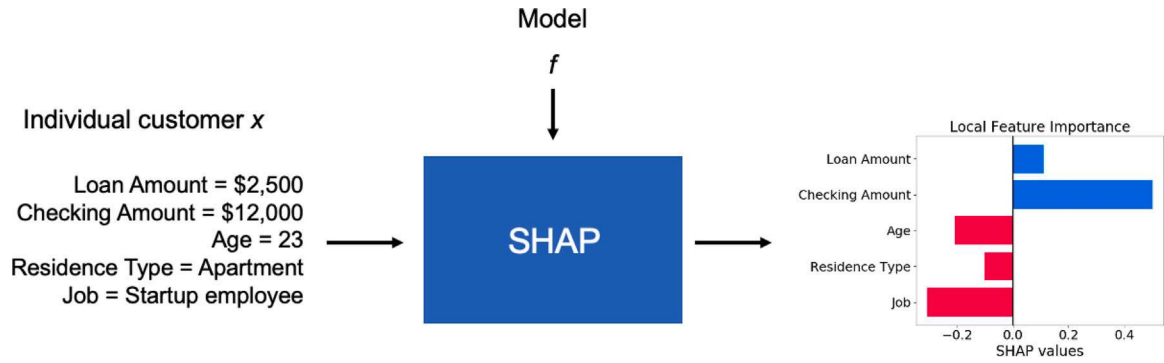
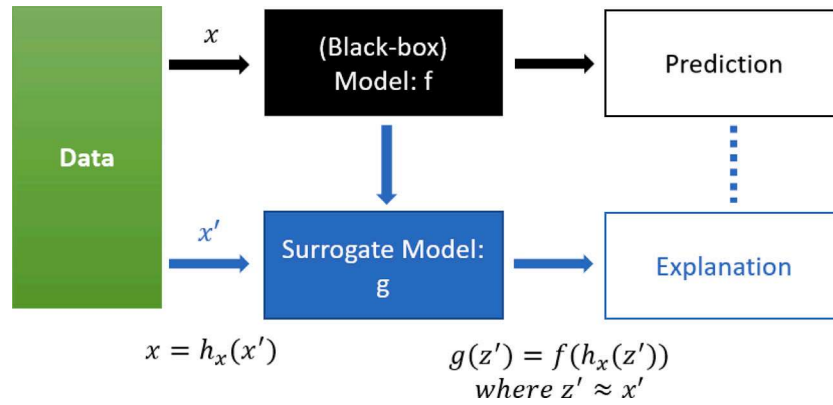
Fig. 1. Simple illustration of SHAP (<https://iancovert.com/>).

Fig. 2. Approximation process of SHAP [25].

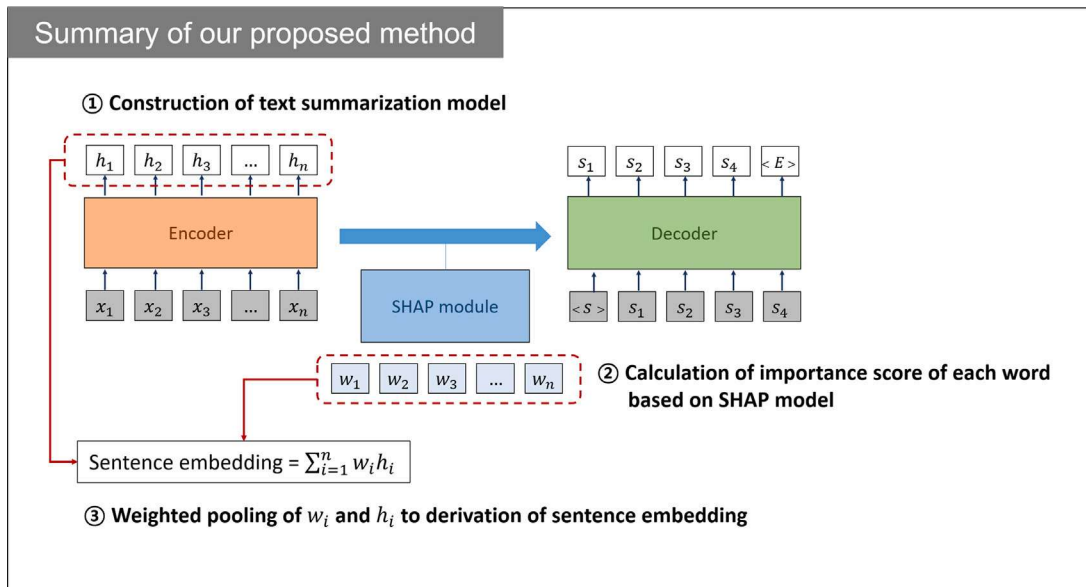


Fig. 3. Summary of our proposed method.

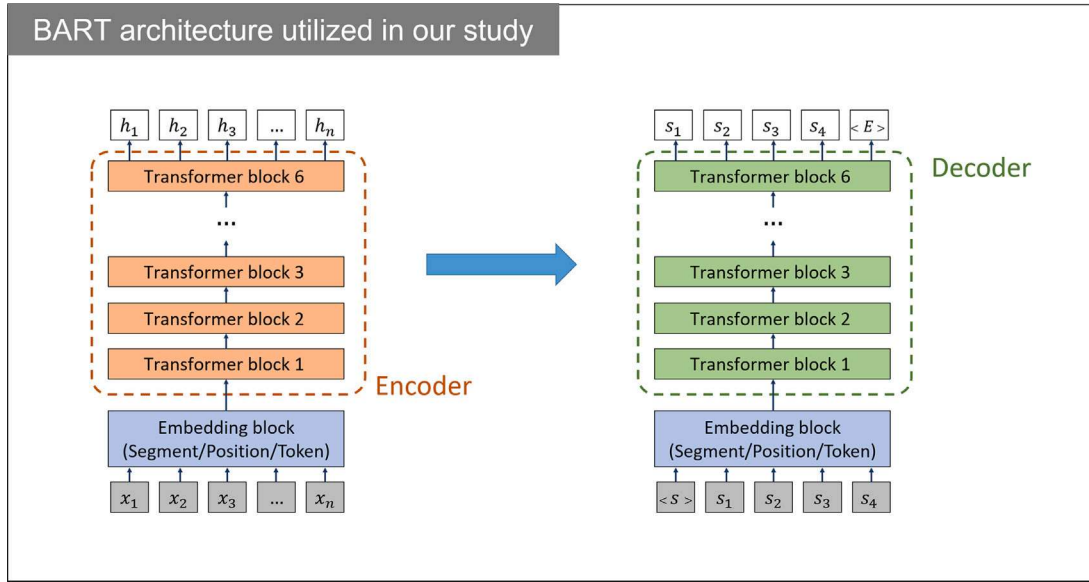


Fig. 4. Architecture of the BART utilized in our study.

3.1. Construction of the text summarization model

We constructed the text summarization model using a pre-trained Bidirectional and Auto-Regressive Transformer (BART_{base}) architecture [32]. BART is one of the most widely used text summary models, and may be implemented as BART_{base} or BART_{arge} according to the number of layers. In the present study, BART was implemented as a sequence-to-sequence model with a bidirectional encoder over corrupted text, and a left-to-right autoregressive decoder.

We used six layers in the encoder and decoder, with six self-attention heads for each layer. Each layer of the decoder also performs cross attention over the final hidden layer of the encoder. The overall architecture of the pre-trained BART model is illustrated in Fig. 4.

Essentially, we utilized the pre-trained BART model developed in [32], which was pre-trained with five tasks (token masking, token deletion, text infilling, sentence permutation, document rotation) and four datasets (Bookcorpus, CC-News, Openwebtext, Stories).

We alternately applied two fine-tuning stages (Fig. 5). The first stage was fine-tuning for dataset adaptation, wherein token masking and deletion were performed with the dataset used for each experiment. The second stage was learning for the summarization task, which was performed using the Extreme Summarization (XSum) dataset.

For fine-tuning, the batch size was set to 64, the learning rate was set to 0.0001, and the model was alternately trained for 20 epochs. In addition, an ablation study was performed to verify the effect of fine-tuning on the dataset, as described in the Experiment section.

3.2. Calculation of token importance using text summarization model

The importance of each token is calculated using the text summarization model described in the previous section. Specifically, the importance of each token $w = (w_1, w_2, \dots, w_n)$ is learned by attaching the SHAP explainer module to the text summarization model.

As mentioned in Section 2, SHAP is the most widespread approach for calculating the sentiment explainability score in classification. However, we combined the SHAP model with the text summarization model to calculate the importance score of each token.

The classification and summarization models employ the same essential approach to calculate the token importance score. Using the

method to obtain the sentiment-explainability score for the output sentiment in the classification model, the importance score of the input text is obtained for each output token of the summarization model.

As mentioned previously, SHAP approximates a simplified input x' and surrogate model g given an original input x and complex black-box model f . The simplified input x' is approximated by the mapping function $x = h_x(x')$. Based on the assumption that $z' \approx x'$, $z \in \{0, 1\}^M$ is represented as a binary variable (1: feature included in the model, 0: feature excluded from the model). We then train an explainable model g such that $g(z') \approx f(h_x(x'))$. The model g is approximated by a linear function of binary variables as shown below, where ϕ_i denotes the importance score of the i th variable:

$$g(z') = \phi_0 + \sum_{i=1}^M \phi_i z'_i$$

In a real implementation, the result of tokenizing the entire dataset becomes the original input x , and the pre-trained and fine-tuned summarization model becomes the black-box model f . By entering x and f into the explainer function, we can approximate the simplified input x' and surrogate model g . Finally, each token's importance score ϕ_i is calculated using the result of the explainer function.

As shown in Figs. 6 and 7, we can use SHAP to derive an importance score for each output token. Subsequently, the importance of each token w_i in the input is calculated as the average importance score for each output token. In detail, assuming that the summarization model's output consists of m tokens, the importance of token w_i is calculated as the average of the importance scores $[w_i^1, w_i^2, \dots, w_i^m]$ for m output tokens.

3.3. Derivation of sentence embeddings reflecting token importance

The sentence embedding is derived using the importance of each token. First, the encoder output $h = (h_1, h_2, \dots, h_n)$ is used as the embedding information for each token. By considering each token's weight $w = (w_1, w_2, \dots, w_n)$ as derived in the previous section, the final sentence embedding is obtained through a weight pooling of the token embedding as follows:

$$\text{sentence embedding} = \sum_{i=1}^n w_i h_i.$$

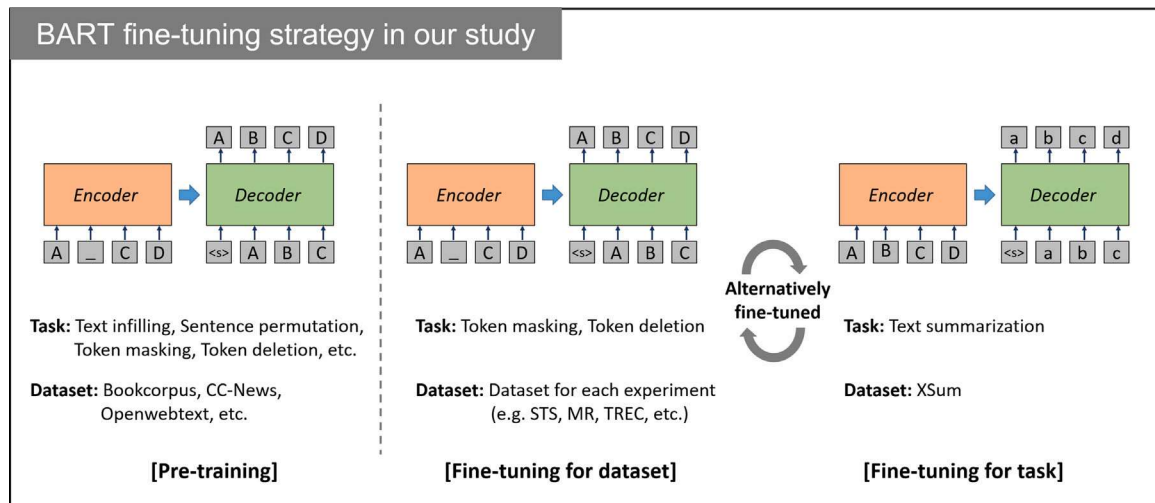
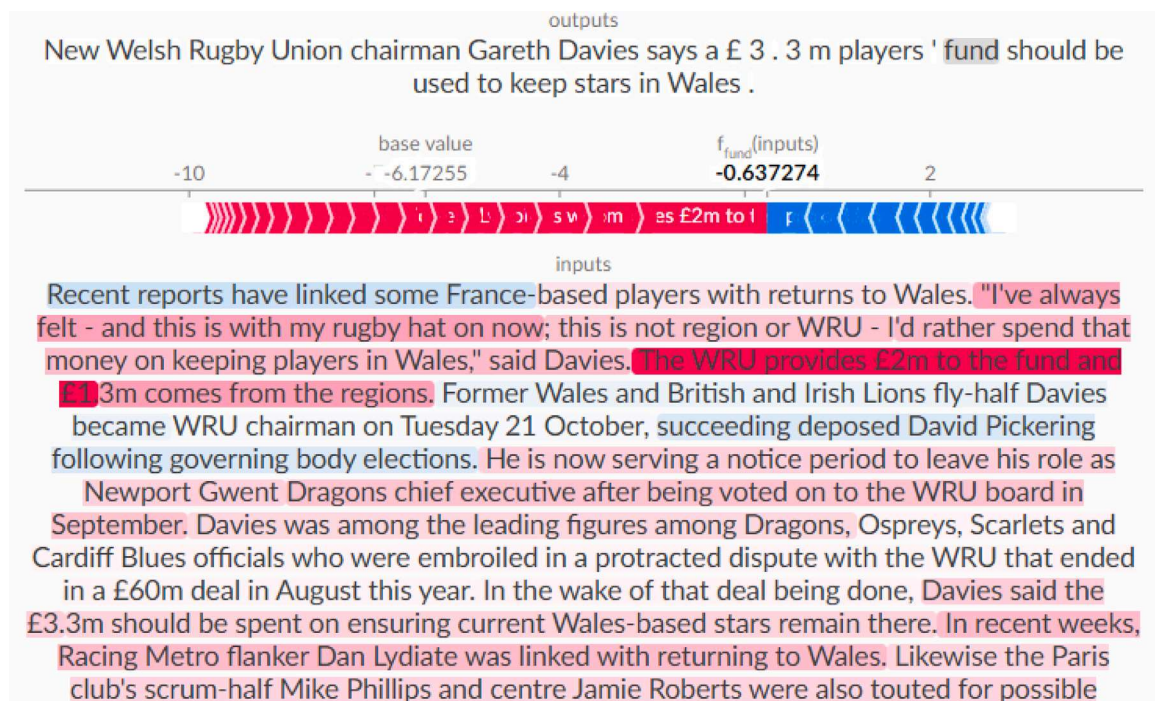


Fig. 5. Fine-tuning strategy in our study.

Fig. 6. Importance score of each input token for the "fund" output token (<https://shap.readthedocs.io/>).

4. Experiments

In this section, we describe the experiment setup and experimental results to validate the effectiveness of the proposed methodology.

4.1. Experiment setup

We evaluated our proposed model on two tasks: (1) semantic textual similarity (STS) and (2) sentiment classification. STS tasks encompass pairs of sentences collected from various sources, where each sentence pair is labeled with a score between 0 and 5 based on semantic relations. We evaluated how well the cosine similarity distance between two sentence embeddings correlates with a human-labeled similarity score using the Pearson and Spearman correlations. We also evaluated

our proposed model on various sentiment classification tasks. Sentence embeddings constructed by each model were used as features to fit a simple classifier such as a logistic regression or naive Bayes classifiers. For most classification tasks, 10-fold validation was applied to compute the test accuracy. A detailed description of this dataset is provided in the next section.

We compared the proposed method's experimental results with those obtained by the most widespread sentence embedding methods as described in Section 2: (1) averaging GloVe embedding; (2) average BERT, RoBERTa and BART embeddings; (3) CLS BERT and RoBERTa embeddings; (4) S-BERT; (5) SimCSE (BERT); (6) ConSERT (BERT); (7) SG-OPT (BERT); (8) WhiteningBERT; (9) PromptBERT; (10) DiffCSE; and (11) Sentence-T5 (Encoder mean). A brief description of each method is provided in Table 1.

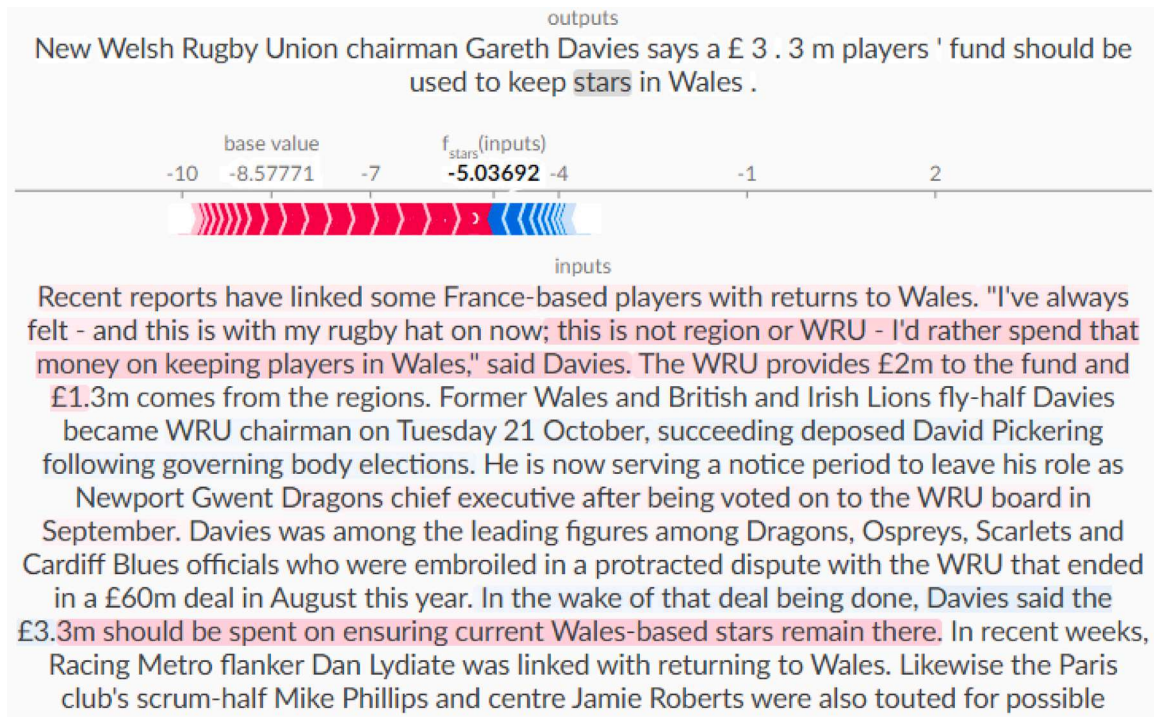


Fig. 7. Importance score of each input token for the “stars” output token (<https://shap.readthedocs.io/>).

Table 1

Brief descriptions of benchmark models.

Benchmarks	Description
GloVe [33]	Averages the GloVe embeddings of words to obtain a sentence embedding
Average BERT/RoBERTa [34]	Adopts the average of token embeddings of the last layer of BERT
Average BART	Adopts the average of token embeddings of the last encoding layer of BART
CLS of BERT/RoBERTa [34]	Adopts the CLS token embeddings of the last layer of BERT
S-BERT [10]	A modification of the pretrained BERT network that uses the Siamese and triplet network structures, which can be compared using cosine similarity
SimCSE [14]	Employs a simple contrastive learning framework that utilizes dropout for data augmentation
ConSERT [35]	Employs various text augmentation strategies for the contrastive learning of sentence representation
SG-OPT [16]	A contrastive learning method with a self-guidance mechanism for sentence embedding
WhiteningBERT [12]	Uses the whitening operation to refine representations and reduce dimensionality
PromptBERT [17]	A prompt-based embedding method that reformulates the sentence embedding task as a fill-in-the-blanks problem
DiffCSE [21]	An instance of equivariant contrastive learning that is sensitive to the difference the between original and augmented sentences
Sentence-T5 [22]	Provides the first exploration of sentence embeddings from text-to-text transformers (T5).

All baseline models followed the implementation details suggested their respective studies, and most implementations utilized the Hugging Face Transformer package (<https://huggingface.co/transformers>). In addition to BART, the X5, BertAbs and RoBERTa-Seq2Seq were tested as text summarization models during the ablation study. The RoBERTa-Seq2Seq has the same structure as BART, but it is a model where both the encoder and decoder are composed using the RoBERTa architecture [36]. The implementation details are summarized in Table 2.

4.2. Data description

As mentioned in the previous section, we performed STS and sentiment classification tasks to evaluate performance. For STS, the STS12–STS16 and STS Benchmark (STSB) datasets were used, with their statistics summarized in Table 3.

For the classification task, the following ten datasets were used: (1) MR: movie review, (2) SST-2: the Stanford Sentiment Treebank, (3) CR: customer reviews, (4) SUBJ: a subjectivity/objectivity dataset, (5)

Table 2
Implementation details of our proposed method.

	BART	T-5	BertSumExtAbs	RoBERTa-seq2seq
Hidden dimension	768	768	768	768
Epoch	10	10	10	10
Learning rate	1e-4	5e-5	5e-5	1e-4
Batch size	128	128	128	128
Number of head	16	12	12	16
Number of layer	12	12	12	12

Table 3
Statistics of STS datasets.

Dataset	$N_{training}$	N_{test}
STS12	0	3108
STS13	0	1500
STS14	0	3750
STS15	0	3000
STS16	0	1186
STSb	5749	1379

TREC: a question type dataset, (6) MPQA: Opinion polarity analysis for news article, (7) MRPC: semantic equivalence analysis, (8) SemEval14 (Restaurant): sentiment analysis for restaurant reviews, (9) SemEval14 (Laptop): sentiment analysis for laptop reviews, and (10) ACL14. The statistics of these datasets are summarized in Table 4.

4.3. Experiment results

In this section, we present the comparative experimental results between the proposed methodology and the baselines, along with qualitative analysis findings. Additionally, we provide details of an ablation study conducted to ascertain the influence of various variables.

4.3.1. Performance comparison

The proposed and benchmark methods are compared in terms of performance on STS tasks in Table 5. As revealed in the figure, our proposed method exhibited the highest performance for three STS datasets (STS12–STS14) and the second-best performance with a narrow margin on three other tasks (STS15–16, and STSb). These results indicate that our proposed method effectively calculates the similarity between sentences based on that between the core tokens in each sentence by reflecting the importance of each token in the sentence to the sentence embedding. As expected, performance was significantly lower in the case of naive BERT average pooling and CLS tokens, even compared to that of traditional GloVe embedding. Although methods that employ contrastive learning performed better than those using naive average pooling and CLS tokens, performance also varied substantially depending on the task, and the absolute performance was also limited. However, not only did our proposed method yield satisfactory absolute performance, but the performance deviation between tasks was also relatively small. In addition, the comparison with average BART confirms the direct effect of our proposed methodology considering word importance.

Table 6 presents the experimental results for the text classification task. Overall, these results are similar to those for the STS task. Our proposed method obtained the highest performance in eight datasets, and the second-highest performance by a narrow margin only in two datasets.

Unlike the results of the STS task, the naive average pooling and CLS token methods obtained higher results than traditional GloVe embedding. Furthermore, although methods using contrastive learning performed better than those based on naive average pooling and CLS tokens, the degree of improvement was small, and the performance of the former was also limited. By contrast, our proposed method outperformed the existing methods with minimal deviation in the performance of different tasks. This proves that our proposed method

effectively reflects both the local information of the token, and the global information by considering the importance of the token in sentence embedding.

4.3.2. Qualitative analysis

The key differentiation of the proposed methodology is that sentence embedding is performed based on the importance score of each token derived from the text summarization model. That is, the objective is to increase the influence of important words the sentence embedding task. Therefore, we further verified how effectively the importance score represents the actual importance of each word.

In Table 7, the five words associated with the highest importance scores are presented for each example sentence. These results indicate that words with substantial importance in each sentence were effectively derived with high scores by the text summarization model. As we expected, the explainability score derived from the model proved to be a meaningful approach to measuring the importance score of each word.

4.3.3. Ablation studies

In this section, we compared classification performance when using the pre-trained BART model with and without dataset fine-tuning for the text summarization model. Subsequently, we performed an ablation study using other text summarization models such as T-5 and BertSumExtAbs.

Table 8 summarizes the results of the ablation study on the classification task, indicating that fine-tuning improves performance when using the text summarization model. This appears to be effective not only for tuning the pre-trained model to fit the new dataset but also for processing new tokens that are not in the token pool of the pre-trained model.

Furthermore, the difference in results when the summarization model is changed or different tasks are conducted is small, further demonstrating the model's robustness.

5. Conclusion

In this study, we propose an advanced sentence embedding method based on a pre-trained language model that reflects the importance of tokens. Although pre-trained language models perform well on many tasks, they utilize naïve approaches – namely CLS tokens or average token pooling – when applied to sentence embedding. Consequently, these models cannot reflect local token information and global sentence information simultaneously, which restricts their performance. Furthermore, although many studies have obtained improvements in performance based on contrastive learning, they had fundamental limitations due to being restricted to the same naïve approaches.

In this study, token importance was derived using a text summarization model, and the sentence embedding method was developed to account for this importance. Specifically, the token importance is calculated by combining the XAI module with a text summarization model, and the final sentence embedding is derived using weighted pooling. Thus, we derive a sentence embedding that considers both the local information of the token embedding, and the global information of the entire sentence.

Experimental results on the STS and text classification tasks reveal that our proposed method not only outperforms existing methods but

Table 4
Statistics for classification datasets.

Dataset	Task	Number of classes	$N_{training}$	N_{test}
MR	Sentiment analysis	2	8530	1066
SST-2	Sentiment analysis	2	7447	1752
CR	Sentiment analysis	2	4082	452
SUBJ	Subjectivity analysis	2	9000	1000
TREC	Question classification prediction	6	5452	500
MPQA	Opinion polarity analysis	2	9000	1000
MRPC	Semantic equivalence analysis	2	4076	1725
SemEval14 (restraunt)	Sentiment analysis	3	3618	1120
SemEval14 (laptop)	Sentiment analysis	3	2328	638
ACL14	Sentiment analysis	3	6248	692

Table 5
Results for STS task.

Method	STS12	STS13	STS14	STS15	STS16	STSb
GloVe	55.14	70.66	59.73	68.25	63.66	58.02
Average BERT	30.87	59.89	47.73	60.29	63.73	47.29
Average RoBERTa	32.11	56.33	45.22	61.34	61.98	54.53
Average BART	31.42	57.72	46.33	61.10	62.49	52.71
CLS of BERT	21.54	32.11	21.28	37.89	44.24	20.30
CLS of RoBERTa	16.67	45.57	30.36	55.08	56.98	45.41
S-BERT	70.73	83.21	75.32	82.41	80.11	7998
SimCSE (BERT)	68.40	82.41	74.38	80.91	78.56	76.85
SimCSE (RoBERTa)	70.16	81.77	73.24	81.36	80.65	80.22
ConSERT	64.64	78.49	69.07	79.72	75.95	73.97
SG-OPT (BERT)	66.84	80.13	71.23	81.56	77.17	77.23
SG-OPT (RoBERTa)	62.57	78.96	69.24	79.99	77.17	77.60
WhiteningBERT	57.83	66.90	60.90	75.08	71.31	68.24
PromptBERT	75.48	85.59	80.57	85.99	81.08	84.56
DiffCSE (BERT)	72.28	84.43	76.47	83.90	80.54	80.59
DiffCSE (RoBERTa)	70.05	83.43	75.49	82.81	82.12	82.38
Sentence-T5 (Enc. mean)	78.05	85.84	82.19	87.46	84.03	86.03
Ours	78.56	86.91	82.41	86.34	83.44	85.17

Table 6
Results of text classification task.

Method	MR	SST-2	CR	SUBJ	TREC	MPQA	MRPC	restraunt	laptop	ACL14
GloVe	77.25	80.18	78.30	91.17	83.00	87.95	72.87	78.91	85.14	83.79
Average BERT	78.66	84.40	86.25	94.37	92.80	88.66	69.54	79.31	85.06	84.83
Average BART	80.14	84.94	87.15	93.06	90.51	87.32	70.66	80.27	84.94	85.59
CLS of BERT	78.68	84.13	84.85	94.21	91.40	88.23	71.13	78.52	85.01	84.46
S-BERT	81.96	87.49	87.55	93.29	89.83	87.97	75.42	81.55	85.12	86.04
SimCSE (BERT)	81.18	85.50	86.46	94.45	89.80	88.88	74.43	81.79	85.41	86.71
SimCSE (RoBERTa)	81.04	86.60	87.74	93.28	84.60	86.94	73.68	80.45	84.76	85.16
SG-OPT (BERT)	82.47	86.20	87.42	95.40	91.60	88.92	74.21	80.77	84.19	85.48
DiffCSE (BERT)	82.69	86.60	87.23	95.23	90.40	89.28	76.58	82.81	86.44	86.81
DiffCSE (RoBERTa)	82.82	88.63	88.61	94.32	90.40	87.81	76.81	82.96	86.49	87.45
Sentence-T5 (Enc. mean)	85.75	91.76	88.74	94.58	96.40	90.95	75.19	84.35	89.89	90.52
Ours	84.78	90.85	89.87	96.14	94.52	90.46	77.84	85.81	91.72	92.15

also performs consistently on different tasks. In addition, results of the ablation study reveal that the proposed method is a robust model with little dependency on the text summarization model.

From the experimental results, we conclude that our proposed method will be an important basis for sentence embedding methods based on token importance in the future. Future research could also extend the scope of sentence-embedding methods by considering the importance of tokens using other approaches. Moreover, such studies are expected to aid in the widespread application of sentence embedding in various tasks arising within the text-mining domain.

CRedit authorship contribution statement

Yuhoo Cha: Methodology, Validation, Software, Data curation, Writing – original draft. **Younghoon Lee:** Conceptualization, Methodology, Writing – original draft, Writing – review & editing, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Table 7

Top 5 words by importance score for each example sentence.

Example sentence	Top 5 words with high importance score
“Very good especially for notebooks with micro hdmi and you want to connect it to a computer screen”	[notebooks, screen, connect, good, hdmi]
“The quality on these speakers is insanely good and doesn’t sound muddy when adjusting bass. Very happy with these.”	[speaker, sound, bass, quality, good]
“Wish I could tilt it more up & down, but it works fine especially considering the price. Great price, great product. we’re going to buy another one for our upstairs TV.”	[TV, price, product, great, fine]
“We bought this for my daughter for her birthday in July. She turned 13. She played it 3 times with the new albums she got for her birthday. She went to listen to some more new albums she bought and it didn’t work. The turntable would not turn!!! I went to replace this item and I was literally 5 days from the last chance date that I had to return this item. Not happy at all. Now she is upset!!!”	[album, work, upset, bought, happy]
“Great pair of headphones for the price. If you’ve ever taken an overnight flight and had trouble sleeping because of the ambient noise, it’s amazing what a little noise reduction can do. You can use these headphones without even plugging them in, just for the noise reduction. On the other hand, they sound pretty good as well. I found these phones comfortable enough to wear for several hours straight. All in all, a good investment in peace and quiet!”	[headphones, price, trouble, noise, sound]
“I am having a hard time getting the nut tight behind the mount for the monitor. It seems tight, but the monitor still pivots left and right way to easy. I also see lots of people complaining whey they tried to get it really tight they broke the mount. I have used better mounts that are only a few dollars more. Spend a little more and get a decent mount.”	[monitor, broke, dollars, easy, complaining]
“I’m really not qualified to review headphones, but some members on head-fi gave them really good reviews and so far they’re the best I’ve had or used. They’re extremely clear, and while somewhat lacking on bass, you still feel and hear it it’s just not overpowering. Listening to these headphones is akin to the experience of eating Dark Chocolate.”	[headphones, clear, bass, listening, experience]
“I was very disappointed with the quality of the headphones they sound very low the battery dies in less then 2 h which I find ironic because they are not even loud. The range from your phone to the actual headset is probably only like 2 ft because it disconnects everytime I got my phone in my pocket all around upset about the product especially for this price I could’ve bought any one of the 30 or 40 dollar headphone and it probably would’ve been the same or better”	[headphones, disappointed, loud, battery, upset]
“Nice sound for what it is – a small Bluetooth speaker. Syncs easily with my iPad. Good clear sound, decent bass. Sounds great on talk radio stations. Nice little unit for the price.”	[sound, speaker, clear, bass, nice]
“Sorry, this cable worked for about a week then broke. I had ordered others before and same result. My fault for thinking things would change. Thought I had gotten a bad set, but that is the rule...not the exception. I’ll try the more expensive ones from Bose.”	[cable, bad, broke, expensive, sorry]

Table 8
Ablation study.

Task	W/dataset fine-tuning	W/O dataset fine-tuning	T-5	BertSumExtAbs	RoBERTa-seq2seq
MR	83.91	82.01	82.49	82.91	83.32
SST-2	89.71	87.15	88.62	88.12	89.71
CR	89.87	87.26	89.10	89.55	90.34
SUBJ	96.14	94.88	95.81	96.01	96.52
TREC	93.52	91.31	93.57	92.97	94.17

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